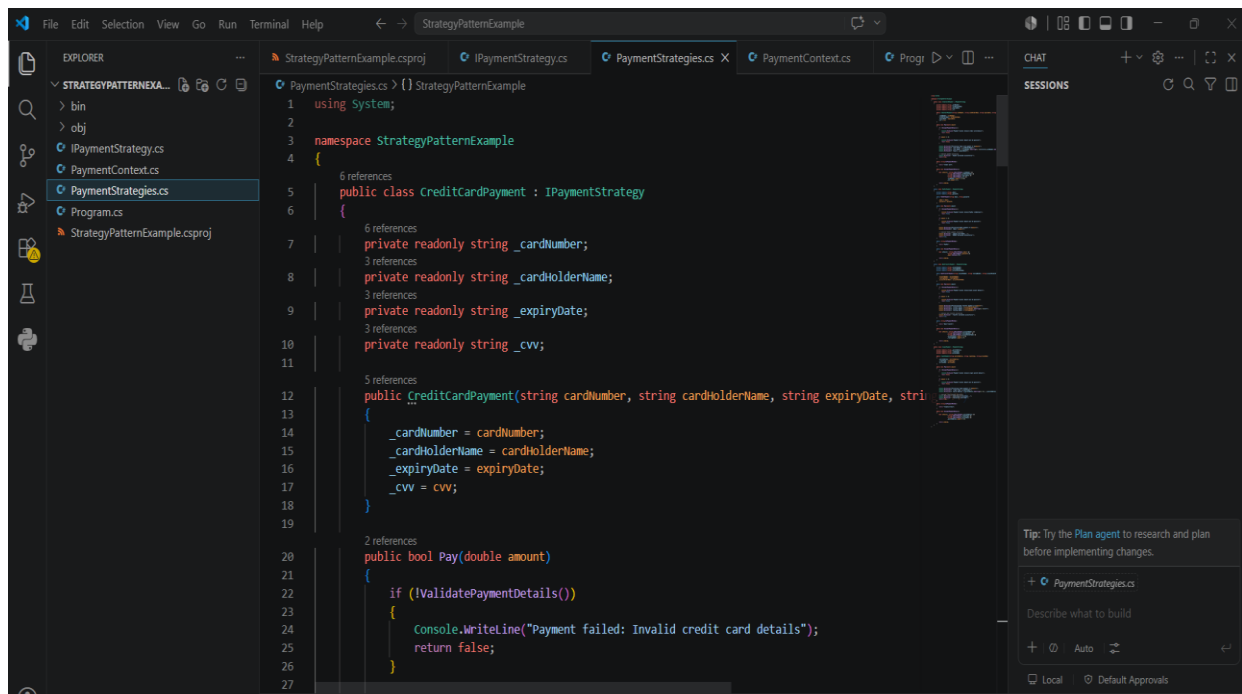


Hands-On

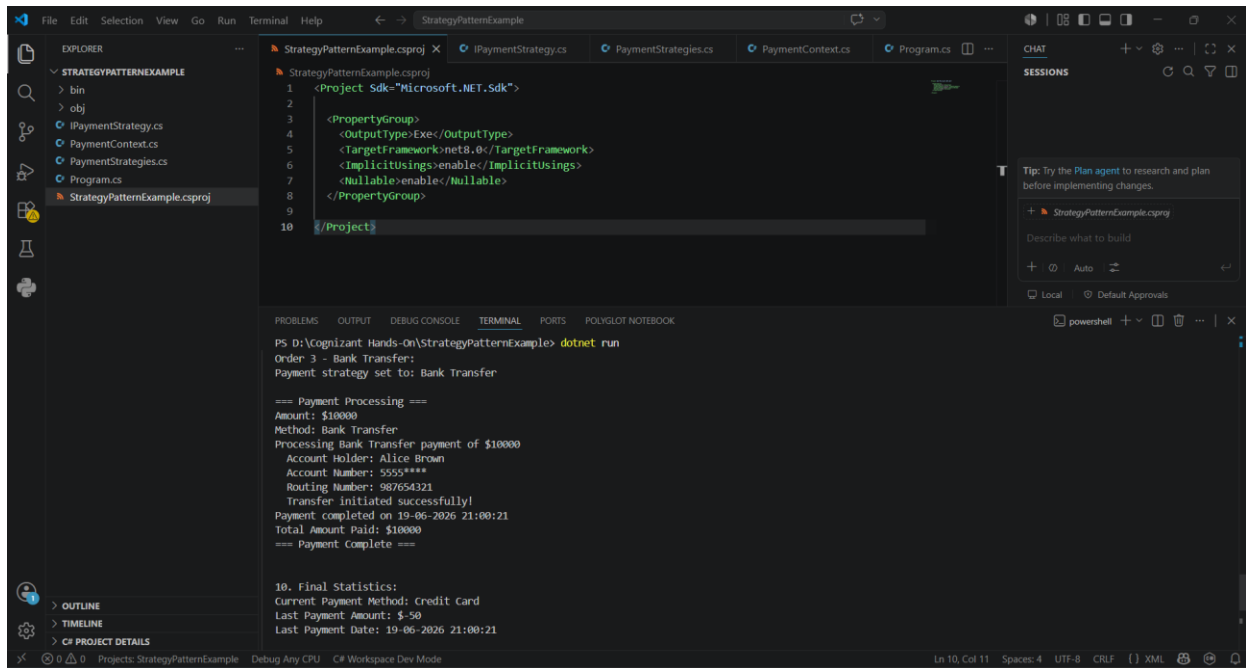
Exercise 8: Implementing the Strategy Pattern

During this module, I was able to implement, execute and test the Strategy Pattern in .NET has developed a project called StrategyPatternExample. I designed a PaymentStrategy interface that defines a Pay() method and concrete strategy classes like CreditCardPayment and PayPalPayment that offer different payment processing strategies. I also created a PaymentContext class that holds a reference to a PaymentStrategy object and performs the chosen payment strategy at runtime. Once implemented, I tested the program with various payment methods by dynamically switching between strategies to ensure that they work. This practical exercise was useful for me to understand the Strategy Pattern, which allows to choose an algorithm or a behavior at runtime, makes it flexible, reduces conditional logic and makes it easy to understand why. More easily extendable and maintainable NET applications.

Coding:



```
1 using System;
2
3 namespace StrategyPatternExample
4 {
5     6 references
6     public class CreditCardPayment : IPaymentStrategy
7     {
8         6 references
9         private readonly string _cardNumber;
10        3 references
11        private readonly string _cardHolderName;
12        3 references
13        private readonly string _expiryDate;
14        3 references
15        private readonly string _cvv;
16
17        5 references
18        public CreditCardPayment(string cardNumber, string cardHolderName, string expiryDate, string cvv)
19        {
20            _cardNumber = cardNumber;
21            _cardHolderName = cardHolderName;
22            _expiryDate = expiryDate;
23            _cvv = cvv;
24        }
25
26        2 references
27        public bool Pay(double amount)
28        {
29            if (!ValidatePaymentDetails())
30            {
31                Console.WriteLine("Payment failed: Invalid credit card details");
32                return false;
33            }
34        }
35    }
36 }
```



Output:

